

GRETN, MB, Canada

WMO: 714410

Lat: 49.0308N

Lon: 97.5600W

Elev: 253

StdP: 98.33

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -30.9	(c) -28.3	(d) -34.7	(e) 0.1	(f) -31.0	(g) -32.1	(h) 0.2	(i) -28.1	(j) 15.6	(k) -15.0	(l) 13.9	(m) -14.4	(n) 3.8	(o) 300	(p) 0.703

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	30.5	21.4	28.8	20.3	27.2	19.3	23.5	28.1	22.1	26.8	20.8	25.3	5.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.0	17.2	26.4	20.5	15.6	25.0	19.1	14.3	23.5	71.8	27.9	66.1	26.8	61.3	25.2	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 13.0	(b) 11.4	(c) 10.2	(e) -34.3	(f) 34.0	(g) 3.5	(h) 1.4	(i) -36.8	(j) 35.0	(k) -38.9	(l) 35.7	(m) -40.9	(n) 36.5	(o) -43.4	(p) 37.5		
			DB													
			WB													
				-34.3	25.7	3.4	1.7	-36.7	26.9	-38.7	27.9	-40.5	28.9	-43.0	30.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.6	-15.3	-13.2	-5.3	4.2	11.5	17.4	19.6	18.7	14.0	5.9	-3.9	-11.7		
	DBStd	13.76	8.12	7.74	7.64	5.89	4.97	3.63	3.03	3.30	4.60	5.05	6.63	7.54		
	HDD10.0	3388	784	649	474	186	43	1	0	0	14	147	418	672		
	HDD18.3	5543	1042	883	732	424	219	59	22	37	143	385	667	930		
	CDD10.0	1043	0	0	1	12	88	222	297	268	134	21	0	0		
	CDD18.3	157	0	0	0	0	6	31	61	46	13	1	0	0		
	CDH23.3	1709	0	0	0	4	103	322	580	524	165	11	0	0		
	CDH26.7	444	0	0	0	0	27	77	142	157	39	2	0	0		
(14) Wind	WSAvg	4.9	5.3	5.2	5.3	5.4	5.5	4.6	3.8	3.8	4.5	5.2	5.2	5.2		
(15) Precipitation	PrecAvg	575	23	20	31	29	78	106	77	68	50	41	29	29		
	PrecMax	714	46	50	86	54	149	240	178	160	116	114	79	49		
	PrecMin	439	8	3	8	6	35	31	27	21	11	6	4	12		
	PrecStd	79	10	10	20	15	30	52	40	43	26	31	21	11		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.7	5.1	15.7	23.4	30.0	32.1	31.8	33.1	30.2	24.6	14.2	5.6		
		MCWB	1.8	3.5	10.2	12.8	16.9	21.6	23.9	21.9	19.9	15.6	9.7	2.5		
	2%	DB	1.3	2.4	10.0	19.9	26.2	28.9	29.9	30.4	27.5	19.9	10.1	2.5		
		MCWB	0.2	1.1	6.9	11.1	15.0	19.4	22.6	21.1	18.4	13.0	6.9	0.6		
	5%	DB	-0.9	0.6	6.3	17.1	23.4	27.0	28.2	28.3	25.1	16.8	7.0	0.4		
		MCWB	-1.7	-0.1	4.2	9.9	13.6	18.3	21.2	20.1	17.1	11.6	4.8	-0.6		
	10%	DB	-3.6	-1.7	3.5	14.2	20.8	25.1	26.6	26.3	22.4	14.1	4.5	-1.5		
		MCWB	-4.2	-2.5	2.1	8.5	13.1	17.9	20.3	18.9	16.1	10.1	2.8	-2.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.1	3.7	11.4	14.4	19.3	24.6	26.2	24.8	22.3	16.7	10.2	2.7		
		MCDB	3.5	5.4	14.6	20.4	25.0	28.7	30.0	29.8	27.7	21.8	12.9	5.2		
	2%	WB	0.3	1.2	6.9	12.1	17.0	21.9	24.0	22.6	20.0	14.1	7.0	0.8		
		MCDB	1.3	2.2	9.7	17.8	23.1	26.1	28.0	28.1	24.6	18.8	9.6	2.1		
	5%	WB	-1.8	-0.1	4.2	10.3	15.5	20.0	22.6	21.3	18.4	12.1	4.9	-0.6		
		MCDB	-1.0	0.7	6.4	15.9	20.3	24.7	26.9	26.2	22.9	16.0	6.7	0.4		
	10%	WB	-4.2	-2.4	2.2	8.7	14.0	18.7	21.2	20.0	16.8	10.1	2.8	-2.2		
		MCDB	-3.6	-1.7	3.4	13.8	19.3	23.3	25.6	24.9	21.4	14.0	4.3	-1.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	9.6	9.0	11.3	12.7	11.3	11.8	13.0	12.4	10.0	8.2	8.3		
		MCDBR	10.6	9.5	11.1	17.0	16.9	14.4	13.6	15.6	16.2	14.9	11.9	10.0		
	5% WB	MCWBR	9.7	8.1	8.3	10.0	8.7	7.3	7.7	7.8	8.7	9.2	8.8	8.7		
		MCWBR	10.5	8.8	10.6	15.3	13.1	11.7	12.6	13.5	13.8	13.7	11.1	9.6		
(40) Clear-Sky Solar Irradiance	taub		0.274	0.299	0.305	0.355	0.371	0.382	0.386	0.382	0.347	0.306	0.274	0.255		
	taud		2.271	2.215	2.392	2.348	2.362	2.358	2.354	2.359	2.451	2.538	2.492	2.393		
	Ebn at Noon		781	843	912	893	889	879	868	855	855	844	799	772		
	Edh at Noon		72	97	97	114	117	118	117	112	92	71	58	56		
(44) All-Sky Solar Radiation	RadAvg		1.47	2.71	3.93	4.93	5.48	5.89	6.27	5.16	3.77	2.28	1.30	1.02		
	RadStd		0.16	0.20	0.36	0.42	0.40	0.45	0.26	0.33	0.27	0.32	0.17	0.12		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	-1.23	-1.58	N/A	N/A	+69	N/A
(47) Regional (30 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page